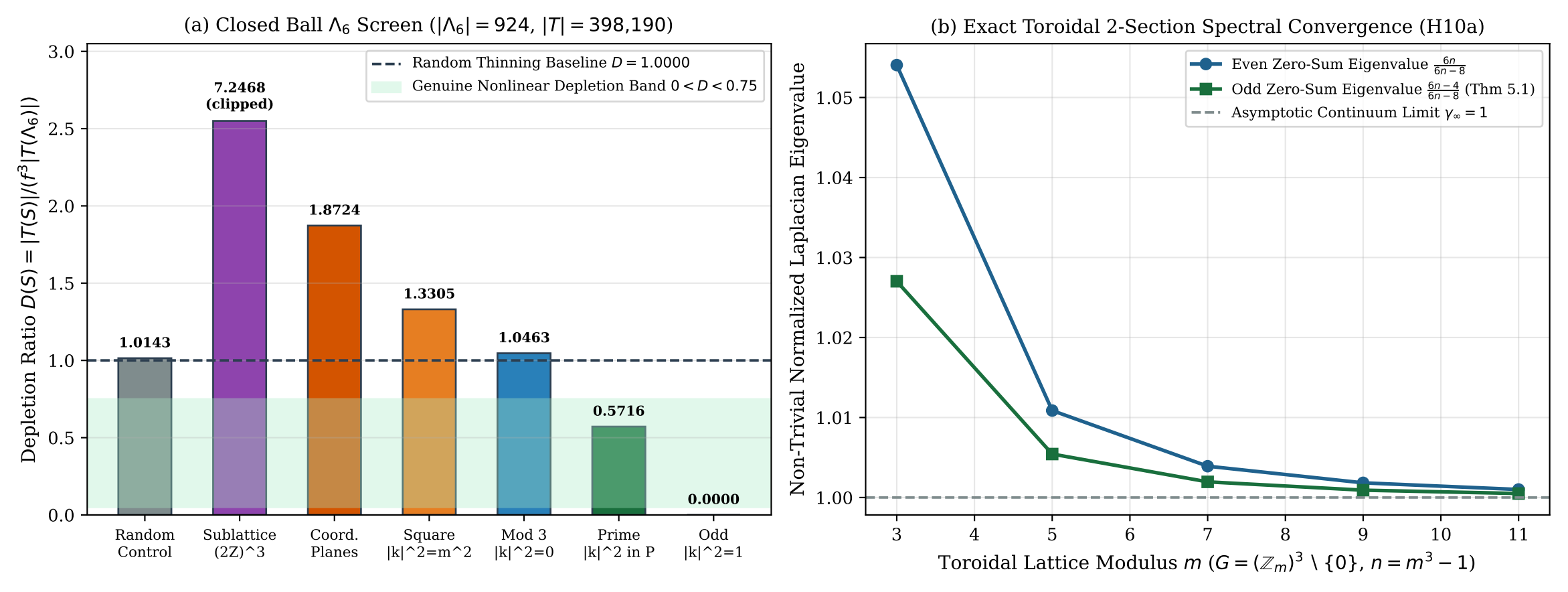
(a) Closed Ball Λ_6 Screen ($|\Lambda_6| = 924$, $|T| = 398{,}190$)
Depletion Ratio $D(S) = |T(S)|/(f^3|T(\Lambda_6)|)$
Random Thinning Baseline $D = 1.0000$
Genuine Nonlinear Depletion Band $0 < D < 0.75$
1.0143
7.2468 (clipped)
1.8724
1.3305
1.0463
0.5716
0.0000
Random Control
Sublattice (2Z)^3
Coord. Planes
Square |k|^2=m^2
Mod 3 |k|^2=0
Prime |k|^2 in P
Odd |k|^2=1
(b) Exact Toroidal 2-Section Spectral Convergence (H10a)
Non-Trivial Normalized Laplacian Eigenvalue
Even Zero-Sum Eigenvalue $\frac{6n}{6n-8}$
Odd Zero-Sum Eigenvalue $\frac{6n-4}{6n-8}$ (Thm 5.1)
Asymptotic Continuum Limit $\gamma_\infty = 1$
Toroidal Lattice Modulus m ($G = (\mathbb{Z}_m)^3 \setminus \{0\}$, $n = m^3 - 1$)